Ian Ruau

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EDUCATION

Auburn University — **Auburn, Alabama, USA**
Doctor of Philosophy in Mathematics — Aug. 2021 - Jul. 2027
Advisors: Dr. Wenxian Shen and Dr. Le Chen

National University of Mar del Plata — **Mar del Plata, Argentina**
Bachelor of Science in Mathematics — Mar. 2014 - Aug. 2020
Advisor: Dr. Horacio De Pasquale

National University of Mar del Plata — **Mar del Plata, Argentina**
Bachelor of Education in Mathematics — Mar. 2010 - Dec. 2015

RESEARCH INTEREST

My research lies in nonlinear partial differential equations (PDEs), dynamical systems, stochastic analysis, and mathematical biology, with a focus on chemotaxis models that describe how populations move in response to chemical signals. In particular, I worked and have been working on the following research projects on a nonlinear parabolic-elliptic system of PDEs that models chemotaxis:

- Global existence and boundedness of solutions
- Persistence and stabilization of solutions
- Bifurcation of positive constant solutions and spike-like solutions
- Stochastic extension of the chemotaxis model

PUBLICATIONS

Submitted

1. Le Chen, Ian Ruau, and Wenxian Shen. "Chemotaxis models with signal-dependent sensitivity and a logistic-type source, III: Bifurcation and spike-like solutions". In: *Preprint arXiv:2610.20976* (Oct. 2026).
2. Le Chen, Ian Ruau, and Wenxian Shen. "Chemotaxis models with signal-dependent sensitivity and a logistic-type source, II: Persistence and stabilization". In: *Preprint arXiv:2604.02599* (Oct. 2026).

Published

1. Le Chen, Ian Ruau, and Wenxian Shen. "Chemotaxis models with signal-dependent sensitivity and a logistic-type source, I: Boundedness and global existence". In: *Journal of Differential Equations.* 476 (2026) 114477.

TEACHING EXPERIENCE

Fall 2026	*Main Instructor*, Actuarial Seminar In Probability MATH-4820, Auburn University
Summer 2026	*Main Instructor*, Calculus III MATH-2630, Auburn University
Spring 2026	*Recitation Leader*, Calculus III MATH-2630, Auburn University
Fall 2025	*Recitation Leader*, Calculus III MATH-2630, Auburn University
Fall 2024	*Recitation Leader*, Calculus III MATH-2630, Auburn University
Spring 2024	*Recitation Leader*, Calculus III MATH-2630, Auburn University
Fall 2023	*Recitation Leader*, Calculus III MATH-2630, Auburn University
Summer 2023	*Main Instructor*, Topics in Linear Algebra MATH-2660, Auburn University
Spring 2023	*Recitation Leader*, Calculus II MATH-1620, Auburn University
Fall 2022	*Recitation Leader*, Calculus II MATH-1620, Auburn University
Summer 2022	*Mathematics Tutor*, Math Tutoring Center, Auburn University
2018 - Spring 2021	*Teaching Assistant*, Algebra B, National University of Mar del Plata
2019 - Spring 2021	*Mathematics Teacher* (5 courses), A. U. Illia National School (CNAI)
2017 - 2018	*Mathematics Teacher* (2 courses), A. U. Illia National School (CNAI)

WORK EXPERIENCE & PROJECTS

Portfolio Comparison under Different Volatility Measurements (Python) Github Webpage 2027

- Project to obtain the certificate of completion of the *Summer 2026 Quant Finance Bootcamp* at *Erdős Institute*: Given a portfolio of options (puts and calls) we compare its performance given different volatility models with the Black-Scholes model.
- We use Black-Scholes historical volatility, EWMA, GARCH(1,1) and EGARCH(1,1) to measure and predict volatility of a given portfolio.
- We analyze the portfolio P&L performance for our volatility models.

Chemotaxis Simulation CLI Tool (Python) Github Webpage 2025

- Python project as part of my ongoing research that simulates a chemotaxis process from a parabollic-elliptic system of differential equations.
- Central differences and the Runge-Kutta numerical method were used to improve the accuracy of the simulations while ghost points were used to treat the Neumann boundary conditions accurately.
- The simulations match theoretical results previously known, and we expect to find new properties of the solutions in our ongoing research.

Simulations on Some Surface Growth Models (Python) Github Webpage 2024

- Project developed for the Applied Stochastic Processes course to simulate a ballistic deposition model.
- We could corroborate through our simulations that the growth model belongs to the universality class such as the one exhibit by the fluctuations of the KPZ equation. Indeed, different ballistic depositions of tetrominoes led to the same universality class of fluctuations.
- Although the empirical results corresponded with the expected behaviour, we could not provide a theoretical proof of the specific region where the fluctuation behaved as the ones of the KPZ equation due to the randomness of the fluctuation paths.

Chicken Behaviour via Markov Chains (Python) 2024

- Project developed for the Applied Stochastic Processes course to analize the behaviour of chickens from a real dataset provided by a Poultry Sciences graduate student.
- By analizing the behavioral data we were able to identify the significant and often ocurring sequential behavioral patterns which led us to find the stationary distribution of the Markov chain associated to the chickens' actions.

Problems of the Sturm-Liouville type associated with Lacunary Series 2020

- Thesis for the B. S. in Mathematics in which a self-adjoint Sturm-Liouville problem contains a coefficient whose Fourier series is a lacunary sequence.
- We proved that the solution of the Sturm-Liouville problem has equally spaced Fourier coefficients given by the lacunary sequence of the problem.

AWARDS

- Summer 2026, *Dr. J. Earl Perry Endowed Summer Graduate Fellowship* recognizing outstanding graduate students for their academic promise and research potential, Auburn University
- Fall 2025, *Haynsworth Fellowship* —Emilie Virginia Haynsworth Memorial Endowment for Excellence in Mathematics—, Auburn University

FUTURE CONFERENCES

Jan. 2027 *AMS 2027 Joint Mathematics Meetings*
Talk title: Chemotaxis models with signal-dependent sensitivity and a logistic-type source: Boundedness and global existence of solutions
Chicago, IL

ATTENDANCE TO CONFERENCES/WORKSHOPS

May 2025 *Third Joint Alabama—Florida Conference on Differential Equations, Dynamical Systems and Applications,* UAB, Birmingham

April 2025 *2025 Southeast Applied and Computational Math Student Workshop,*
Auburn University, Auburn

Nov. 2024 *KU Probability and Statistics Conference 2024 on Stochastic Analysis and Related Areas,*
University of Kansas, Lawrence

May 2023 *Joint Alabama–Florida Conference on Differential Equations, Dynamical Systems and Applications*, Auburn University, Auburn

Dec. 2017 *V International Symposium on Nonlinear PDEs & Free Boundary Problems*
Faculty of Exact and Natural Sciences of the National University of Buenos Aires

FUNDING

May 2025 Travel award to attend the *AMS 2027 Joint Mathematics Meetings* in Chicago, IL, granted by the College of Sciences and Mathematics of Auburn University

May 2025 Travel funds to attend the *Third Joint Alabama—Florida Conference on Differential Equations, Dynamical System and Applications,* granted by UAB, Birmingham

Nov. 2024 Travel funds to attend the *KU Probability and Statistics Conference 2024 on Stochastic Analysis and Related Areas,* granted by the University of Kansas, Lawrence

Dec. 2017 Travel funds to attend the *V International Symposium on Nonlinear PDEs & Free Boundary Problems* granted by the National University of Buenos Aires

OUTREACH

- Fall 2026 – Spring 2027, Senator for the Mathematics and Statistics Graduate Student Council
- Fall 2026, Mathematics Tutor at the Veterans Resource Center, Auburn University
- Spring 2026 Mathematical Puzzle Programs (MaPP) Volunteer, Puzzle Master

- Spring 2025, Mentor for 1st and 2nd year graduate students at Auburn University
- 2025 Science Olympiad Volunteer, Helicopter Division B
- 2024 Science Olympiad Volunteer, Flight Division B
- 2023 Science Olympiad Volunteer, Flight Division C

CERTIFICATES

- *Completion of Quant Finance Bootcamp*, granted by *Erdős Institute*, Summer 2026
- *Certificate to Teach in Engaged and Active Student Learning Classrooms on Auburn University's campus*, granted by *Biggio Center for Enhancement of Teaching and Learning*, Auburn University, Summer 2022

TECHNICAL SKILLS

- Programming in Python and MATLAB
- Microsoft Office Suite (Excel, Word, PowerPoint)
- User of Linux Systems (Manjaro Linux, and Ubuntu)
- Version control through Git/GitHub user

LANGUAGES SKILLS

- Spanish: Native language
- English: Professional user
- French: Independent user (B2 in CEFR)